Transfer Learning with Heterogeneous Feature Spaces in Linear Regression

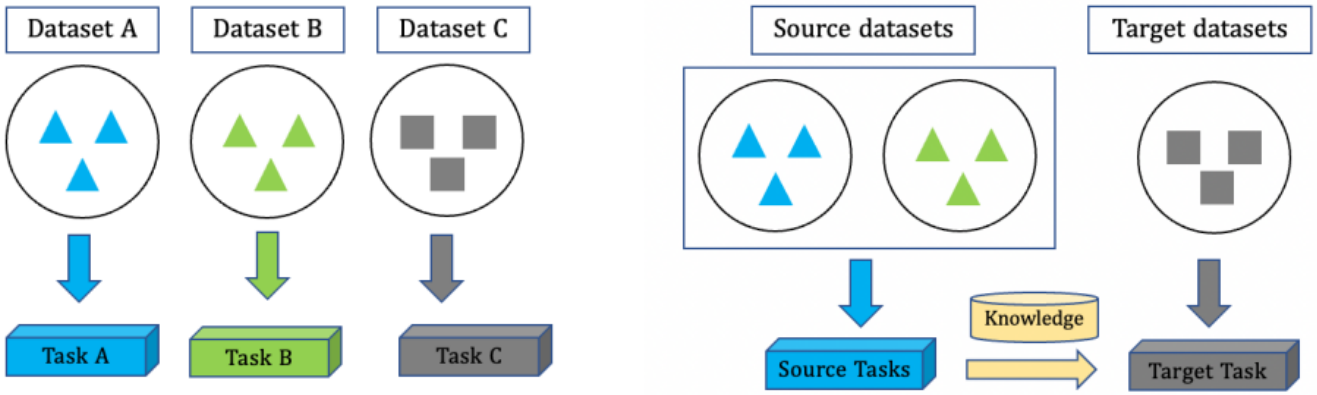
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Background: Transfer Learning

- **Classical Machine Learning:** uses only data from the **target** population.
- **Transfer Learning:** leverages auxiliary information from **source** domains to improve inference for the target problem. (Torrey and Shavlik, 2010; Weiss et al., 2016)



(a) Traditional Machine Learning (b) Transfer Learning

Figure: Classical learning uses target data only; transfer learning borrows source information.

Literature Review

Statistical transfer learning has been extensively studied under the **homogeneous feature-space** assumption.

- **High-dimensional linear models**

  Bastani (2021); Li et al. (2022); Craig et al. (2025); Gu et al. (2025); Zhao et al. (2026).

- **GLM**

  Tian and Feng (2023); Li et al. (2024); Zhang et al. (2025); Zheng et al. (2025).

- **Matrix estimation, multi-response, piecewise-constant**

  Hu and Zhang (2023); Park et al. (2025); Li and Zhang (2025); Wang and Yu (2025).

- **Quantile regression**

  Jin et al. (2024); Zhang and Zhu (2024).

- **Nonparametric regression/classification**

  Cai and Wei (2021); Kpotufe and Martinet (2021); Reeve et al. (2021); Ma et al. (2023); Maity et al. (2024); Fan et al. (2025).

- **GGM and functional data**

  Li et al. (2023); Cai et al. (2024); Wu et al. (2024); Qin et al. (2025).

Background: Heterogeneous Feature Spaces (HFS)

Problem Definition

Reality: different domains often have **different feature sets**.

- Target: the full feature set Θ_0.
- Source: only partial features Θ_k.

Challenge: many existing methods assume $\Theta_0 \equiv \Theta_k$, which breaks down here.

Biomedicine (EHR)

Hospitals often use incompatible recording systems.

- **Target:** full biomarker panel.
- **Source:** only routine measurements.

(Wang et al., 2022)

Computer Vision

Images can differ in resolution or modality.

- **Target:** high-resolution images (1024×1024).
- **Source:** low-resolution images (64×64).

(Duan et al., 2012)

Problem Setup

- **Target Feature Set:** the target domain observes the complete feature set:

$$\Theta_0 = \{1, \ldots, p\}$$

- **Source Feature Sets:** for $k = 1, \ldots, K$, source k observes only a subset:

$$\Theta_k \subseteq \Theta_0$$

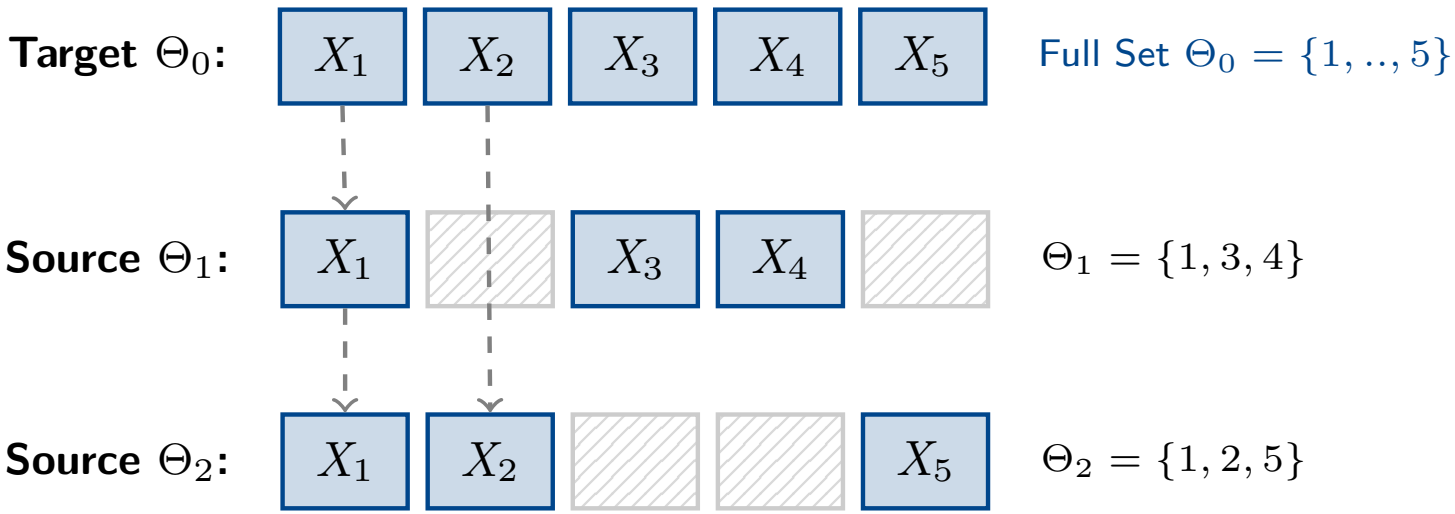


Figure: Each source observes only part of the target feature set.

Related Work

Zhao et al. (2023)

Setting: a single source ($K = 1$) with **partially overlapping features** ($\Theta_1 \neq \Theta_0$).
Data: source summary statistics and target individual-level data.
Limitation:

- Requires **identical marginal models** on the shared covariates ($\Theta_0 \cap \Theta_1$).
- *This is often too restrictive in practice.*

Chang et al. (2024)

Setting: the target feature set is a subset of the source feature set: $\Theta_0 \subsetneq \Theta_k$ (via imputation).
Data: full individual-level data.
Limitation:

- Requires **sufficiently close feature maps** across source and target domains.
- *This can be hard to verify in practice.*

Shared limitations

- ☞ **Risk of Negative Transfer:** Both approaches lack a principled mechanism for **source or sample selection**, so negative transfer can occur.
- ☞ **No entrywise theory:** existing HFS transfer results mainly provide global error bounds and do not capture coordinate-specific effective sample sizes.

Model Setup: From Homogeneous to Heterogeneous

1. Underlying full model

Assume both the target ($k = 0$) and the sources ($k \in [K]$) arise from a full linear model with p covariates:

$$y_i^{(k)} = (\boldsymbol{x}_i^{(k)})^\top \boldsymbol{\beta}^{(k)} + \epsilon_i^{(k)}, \quad i = 1, \ldots, n_k.$$

Here, $\boldsymbol{x}_i^{(k)} \in \mathbb{R}^p$ is the complete feature vector and $\boldsymbol{\Sigma}^{(k)} = \mathrm{cov}(\boldsymbol{x}_i^{(k)})$.

2. Observed heterogeneous feature spaces

- For source k, we observe only the subvector $\tilde{\boldsymbol{x}}_i^{(k)} = (x_{ij}^{(k)})_{j \in \Theta_k} \in \mathbb{R}^{p_k}$.
- The observed source data follow the reduced model:

$$y_i^{(k)} = (\tilde{\boldsymbol{x}}_i^{(k)})^\top \boldsymbol{\theta}^{(k)} + \tilde{\epsilon}_i^{(k)}, \quad k \in [K].$$

- $\boldsymbol{\theta}^{(k)} \in \mathbb{R}^{p_k}$ generally differs from $\boldsymbol{\beta}^{(k)}$.

Projection-Based Imputation

Lift source features to the target space

$$\tilde{\boldsymbol{x}}^{(k)} \in \mathbb{R}^{p_k} \quad \xrightarrow{\boldsymbol{G}^{(k)} \in \mathbb{R}^{p_k \times p}} \quad \boldsymbol{x}_{\text{im}}^{(k)} = \left(\boldsymbol{G}^{(k)}\right)^{\top} \tilde{\boldsymbol{x}}^{(k)} \in \mathbb{R}^{p}.$$

We do **not** require exact recovery of every missing covariate. Bias will be handled by weighting and selection.

What matters for transfer

The projection is evaluated through its target-oriented discrepancy:

$$h_k = \|\boldsymbol{\theta}^{(k)} - \boldsymbol{G}^{(k)} \boldsymbol{\beta}^{(0)}\|_1.$$

A good $\boldsymbol{G}^{(k)}$ makes the projected target coefficient close to the reduced source coefficient.

Why Standard Imputation Is Not Enough

Classical imputation target

If the full source covariance were available, a natural linear imputation matrix would be

$$(\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\Theta_k})^{-1}\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\bullet}.$$

This targets prediction of the unobserved source covariates.

Two HFS obstacles

- **Not identifiable from source k:** covariates outside Θ_k are unobserved, so $\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\bullet}$ is not available.
- **Not target-oriented:** predicting missing covariates need not minimize the error for estimating $\boldsymbol{\beta}^{(0)}$.

Therefore, $\boldsymbol{G}^{(k)}$ is treated as a tunning parameter chosen for transfer, not as a pure missing-covariate reconstruction rule.

Addressing Bias via Importance Weighting

HFS importance weight

After projection, source k is reweighted by

$$\omega^{(k)} = \frac{f_\nu\left(y^{(k)} - (\tilde{\boldsymbol{x}}^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta}^{(0)}\right)}{f_k\left(y^{(k)} - (\tilde{\boldsymbol{x}}^{(k)})^\top \boldsymbol{\theta}^{(k)} \mid \tilde{\boldsymbol{x}}^{(k)}\right)}$$

Here f_ν is a user-specified symmetric density with $\mathbb{E}(\nu) = 0$.

Under HFS, the reduced-model error $\tilde{\epsilon}^{(k)}$ generally depends on $\tilde{\boldsymbol{x}}^{(k)}$. The weight formulation therefore uses the conditional density $f_k\{y^{(k)} - (\tilde{\boldsymbol{x}}^{(k)})^\top \boldsymbol{\theta}^{(k)} \mid \tilde{\boldsymbol{x}}^{(k)}\}$ in the denominator.

Proposition: Population Validity

A convenient sufficient setup

$[p] = \cup_{k=1}^{K} \Theta_k$ and we take $\boldsymbol{G}^{(k)} = P_{\boldsymbol{B}}^{(k)} = (\boldsymbol{B}_{\Theta_k,\Theta_k})^{-1} \boldsymbol{B}_{\Theta_k,\bullet}$ for any positive-definite working matrix $\boldsymbol{B}$.

Under this setup, then

$$\boldsymbol{\beta}^{(0)} = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\operatorname{argmin}} \sum_{k=0}^{K} \alpha_k \, \mathbb{E}_{\tilde{\boldsymbol{z}}^{(k)}} \left[\omega^{(k)} \{ y^{(k)} - (\tilde{\boldsymbol{x}}^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta} \}^2 \right].$$

The setup above is not essential. It suffices that the mixed covariance matrix of target and imputed source covariates

$$\boldsymbol{\Gamma} = \alpha_0 \mathbb{E}[\boldsymbol{x}^{(0)} (\boldsymbol{x}^{(0)})^\top] + \sum_{k=1}^{K} \alpha_k \mathbb{E}[\boldsymbol{x}_{\text{im}}^{(k)} (\boldsymbol{x}_{\text{im}}^{(k)})^\top]$$

is invertible.

Proposition: Weight as Classifier Odds

- Directly estimating $f_k(\tilde{\epsilon}^{(k)} \mid \tilde{\boldsymbol{x}}^{(k)})$ is difficult, especially when p_k is large.
- We instead recover the weight by a binary classification problem.

Synthetic and observed distributions

Let P_k be the distribution of $\tilde{\boldsymbol{z}}^{(k)} = (y^{(k)}, (\tilde{\boldsymbol{x}}^{(k)})^\top)$ and let P_{k0} be the synthetic distribution generated by

$$y^{(k0)} = (\tilde{\boldsymbol{x}}^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta}^{(0)} + \nu.$$

For a probabilistic classifier $D(\tilde{\boldsymbol{z}}) \in (0, 1)$, define

$$\mathcal{R}(D) = -\mathbb{E}_{P_{k0}} \log D(\tilde{\boldsymbol{z}}) - \mathbb{E}_{P_k} \log\{1 - D(\tilde{\boldsymbol{z}})\}.$$

The minimizer $D^{(k)}$ satisfies

$$\frac{D^{(k)}(\tilde{\boldsymbol{z}}^{(k)})}{1 - D^{(k)}(\tilde{\boldsymbol{z}}^{(k)})} = \omega^{(k)}.$$

Classification-Based Weight Estimation

Synthetic-vs.-observed classification

Given initial estimators $\boldsymbol{\beta}_{\text{in}}^{(0)}$ and $\boldsymbol{\theta}_{\text{in}}^{(k)}$ by OLS in low dimension or Lasso in high dimension, we construct a binary classification problem for each source k:

1. Generate pseudo-responses $y_i^{(k0)} = (\tilde{\boldsymbol{x}}_i^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta}_{\text{in}}^{(0)} + \nu_i$ and label them as $L = 1$.
2. Label observed source samples $(y_i^{(k)}, \tilde{\boldsymbol{x}}_i^{(k)})$ as $L = 0$.
3. Train a probabilistic classifier $\hat{D}^{(k)}(y, \tilde{\boldsymbol{x}})$; use logistic regression in low dimension and regularized logistic regression in high dimension.

Classifier odds estimate the importance weight:

$$\hat{\omega}_i^{(k)} = \frac{\hat{D}^{(k)}(\tilde{\boldsymbol{z}}_i^{(k)})}{1 - \hat{D}^{(k)}(\tilde{\boldsymbol{z}}_i^{(k)})}.$$

Construction of the HIW Estimator

Step 1: Empirical weights and sample selection

Given initial estimators, compute $\hat{\eta}_i^{(k)} = (\tilde{\boldsymbol{x}}_i^{(k)})^\top (\boldsymbol{G}^{(k)} \boldsymbol{\beta}_{\text{in}}^{(0)} - \boldsymbol{\theta}_{\text{in}}^{(k)})$ and construct:

Weight: $\hat{\omega}_i^{(k)} = \dfrac{\hat{D}^{(k)}(\tilde{\boldsymbol{z}}_i^{(k)})}{1 - \hat{D}^{(k)}(\tilde{\boldsymbol{z}}_i^{(k)})}$

Subset: $\widehat{\mathcal{I}}_k = \left\{ i \in [n_k] : |y_i^{(k)} - (\tilde{\boldsymbol{x}}_i^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta}_{\text{in}}^{(0)}| \le R, \ \ |\hat{\eta}_i^{(k)}| \le M_k \right\}$

Step 2: The penalized HIW optimization problem

$$\widehat{\boldsymbol{\beta}}_{\text{hiw}}^{(0)} = \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\operatorname{argmin}} \left\{ \frac{1}{2N} \sum_{k=0}^{K} \sum_{i=1}^{n_k} \underbrace{\hat{\omega}_i^{(k)} \mathbb{I}(i \in \widehat{\mathcal{I}}_k)}_{\text{Robust Weighting}} \left(y_i^{(k)} - (\tilde{\boldsymbol{x}}_i^{(k)})^\top \boldsymbol{G}^{(k)} \boldsymbol{\beta} \right)^2 + \sum_{j=1}^{p} \lambda_j |\beta_j| \right\}$$

Conditions for Theory I

Condition 1.

(a) The vectors $\boldsymbol{x}^{(0)}$ and $\tilde{\boldsymbol{x}}^{(k)}$ are sub-Gaussian random vectors, and the errors $\epsilon^{(0)}$ and $\tilde{\epsilon}^{(k)}$ are sub-Gaussian random variables, for all $k \in [K]$.

(b) The smallest and largest eigenvalues of $\boldsymbol{\Sigma}^{(0)}$ and $\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\Theta_k}$ are bounded away from zero and infinity for all $k \in [K]$.

Condition 2.

(a) For each $k \in [K]$, the conditional density $f_k(t \mid \tilde{\boldsymbol{x}}^{(k)})$ is upper bounded and positive for all finite t.

(b) The random variable ν is independent of the source data and satisfies $\mathbb{E}(\nu) = 0$. Its density function $f_\nu(t)$ is symmetric, bounded, and positive for all finite t.

Condition 3.

(a) For $k \in [K]$, the initial estimators $\boldsymbol{\beta}^{(0)}_{\text{in}}$, $\boldsymbol{\theta}^{(k)}_{\text{in}}$ and $\hat{D}^{(k)}(\tilde{\boldsymbol{z}}^{(k)})$ are consistent and have convergence rates such that

$$\|\boldsymbol{\beta}^{(0)}_{\text{in}} - \boldsymbol{\beta}^{(0)}\|_1 = O_p(\gamma_0), \quad \|\boldsymbol{\theta}^{(k)}_{\text{in}} - \boldsymbol{\theta}^{(k)}\|_1 = O_p(\gamma_k),$$

$$\sup_{\tilde{\boldsymbol{z}}^{(k)}} |\hat{D}^{(k)}(\tilde{\boldsymbol{z}}^{(k)}) - \tilde{D}^{(k)}(\tilde{\boldsymbol{z}}^{(k)})| = O_p(q_k).$$

(b) These initial estimators and the projection matrices $\boldsymbol{G}^{(k)}$ are independent of the data used in the transfer learning estimator.

Conditions for Theory II

Define

$$\Phi_k = \{i \in [n_k] : |\eta_i^{(k)}| \le M_k\}, \qquad \Phi_k^{\pm} = \{i \in [n_k] : |\eta_i^{(k)}| \le M_k \pm \delta_n\}.$$

$$\boldsymbol{\Sigma}^{\mathrm{sel}} = \alpha_0 \mathbb{E}[\boldsymbol{x}^{(0)} (\boldsymbol{x}^{(0)})^\top] + \sum_{k=1}^{K} \alpha_k \mathbb{E}[\boldsymbol{x}_{\mathrm{im},i}^{(k)} (\boldsymbol{x}_{\mathrm{im},i}^{(k)})^\top \mathbb{I}(i \in \Phi_k)].$$

Condition 4. Assume that (a) $\max_{j \in \mathcal{S}_0} \sigma_{j,j}^{\mathrm{sel}}$ is upper bounded and that the smallest eigenvalue of $\boldsymbol{\Sigma}_{\mathcal{S}_0,\mathcal{S}_0}^{\mathrm{sel}}$ is bounded away from zero; (b) the same condition holds when $\boldsymbol{\Sigma}^{\mathrm{sel}}$ is defined by replacing Φ_k with either Φ_k^- or Φ_k^+.

Condition 5. For each $k \in [K]$, (a) $\boldsymbol{G}_{\bullet,\Theta_k}^{(k)} = \boldsymbol{I}_{p_k}$, where $\boldsymbol{I}_{p_k} \in \mathbb{R}^{p_k \times p_k}$ is the identity matrix, and (b) $|||\boldsymbol{G}^{(k)}|||_1$ is bounded.

Condition 6. Assume that

$$\max_{j \in \mathcal{S}_0^c} \|(\widehat{\mathcal{T}}_{\mathcal{S}_0,\mathcal{S}_0})^{-1} \widehat{\mathcal{T}}_{\mathcal{S}_0,j}\|_1 \le \iota$$

for some constant $0 \le \iota < 1$.

Condition 7. There exists a positive constant ι_1 such that, for every $j \in [s_0]$,

$$\sum_{l \in [s_0]} \left| [(\widehat{\mathcal{T}}_{\mathcal{S}_0,\mathcal{S}_0})^{-1}]_{j,l} \right| \lambda_{\mathcal{S}_0,l} \le \iota_1 [(\widehat{\mathcal{T}}_{\mathcal{S}_0,\mathcal{S}_0})^{-1}]_{j,j} \lambda_{\mathcal{S}_0,j}.$$

Coordinate-Specific Effective Sample Size

Definition

Let $\Xi_j = \{k \in [K] : j \in \Theta_k\}$. For coordinate j,

$$N_j = n_0 + N_j^{\mathrm{dir}} + N_j^{\mathrm{imp}},$$

where

$$N_j^{\mathrm{dir}} = \sum_{k \in \Xi_j} \mathbb{E}(n_{\mathcal{I}_k}) [\boldsymbol{\Sigma}_{\mathcal{I}}^{(k)}]_{j,j},$$

$$N_j^{\mathrm{imp}} = \sum_{k \notin \Xi_j} \mathbb{E}(n_{\mathcal{I}_k}) (\boldsymbol{G}_{\bullet,j}^{(k)})^\top \boldsymbol{\Sigma}_{\mathcal{I}}^{(k)} \boldsymbol{G}_{\bullet,j}^{(k)},$$

$$\boldsymbol{\Sigma}_{\mathcal{I}}^{(k)} = \mathbb{E}[\tilde{\boldsymbol{x}}_i^{(k)} (\tilde{\boldsymbol{x}}_i^{(k)})^\top \mid i \in \mathcal{I}_k].$$

- N_j^{dir} comes from sources where X_j is observed;
- N_j^{imp} comes from sources where X_j is supplied through projection

Low-Dimensional Convergence Rates

Theorem 1: entry-wise rate

Under Conditions 1–5, with $p \ll \min_k n_k$, the unpenalized HIW estimator satisfies

$$(\hat{\beta}^{(0)}_{\mathrm{hiw},j} - \beta^{(0)}_j)^2 = O_p\left(\frac{1}{N_j}(1+g)\right), \qquad j \in [p].$$

The corresponding global rate is

$$\|\widehat{\boldsymbol{\beta}}^{(0)}_{\mathrm{hiw}} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\sum_{j=1}^{p} \frac{1}{N_j}(1+g)\right).$$

Interpreting the Bias Term g

The term g quantifies the cost of estimating weights and subsets:

$$g = \sum_{k=1}^{K} \mathbb{E}(n_{\mathcal{I}_k}) \left[\gamma_0 \wedge (M_k + h_k)\right]^2$$

- γ_0 is the initial target-estimator rate.
- $h_k = \|\boldsymbol{\theta}^{(k)} - \boldsymbol{G}^{(k)}\boldsymbol{\beta}^{(0)}\|_1$: **Projection Discrepancy**.

The quantity h_k measures the gap between the *source marginal parameter* $\boldsymbol{\theta}^{(k)}$ and the *projected target parameter* $\boldsymbol{G}^{(k)}\boldsymbol{\beta}^{(0)}$.

- Even when $\|\boldsymbol{\beta}^{(k)} - \boldsymbol{\beta}^{(0)}\|_1$ is large, an appropriate projection $\boldsymbol{G}^{(k)}$ can still make $h_k \approx 0$.
- **Contrast with Homogeneous Case:** standard transfer learning typically requires $\boldsymbol{\beta}^{(k)} \approx \boldsymbol{\beta}^{(0)}$.

High-Dimensional Global Convergence

Theorem 2: global convergence

Under Conditions 1–6, the penalized HIW estimator satisfies

$$\|\widehat{\boldsymbol{\beta}}^{(0)}_{\mathrm{hiw}} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\rho_m^2 \frac{s_0 \log s_0}{N_{\min,\mathcal{S}_0}}(1+g)\right),$$

Effective-sample imbalance

$$N_{\min,\mathcal{S}_0} = \min_{j\in\mathcal{S}_0} N_j, \qquad N_{\max,\mathcal{S}_0} = \max_{j\in\mathcal{S}_0} N_j, \qquad \rho_m = \frac{N_{\max,\mathcal{S}_0}}{N_{\min,\mathcal{S}_0}}.$$

If active-coordinate sample sizes are comparable, then $\rho_m = O(1)$.

High-Dimensional Entry-Wise Rates

Theorem 3: entry-wise convergence

Let $\widehat{\mathcal{S}}_0 = \{j : \hat{\beta}^{(0)}_{\mathrm{hiw},j} \neq 0\}$. Under Conditions 1–7, with $s_0 \ll \min_k n_k$, the penalized HIW estimator satisfies

$$\widehat{\mathcal{S}}_0 \subseteq \mathcal{S}_0, \qquad (\hat{\beta}^{(0)}_{\mathrm{hiw},j} - \beta^{(0)}_j)^2 = O_p\left(\frac{\log s_0}{N_j}(1+g)\right), \quad j \in \mathcal{S}_0.$$

Hence, the global rate is

$$\|\widehat{\boldsymbol{\beta}}^{(0)}_{\mathrm{hiw}} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\sum_{j\in\mathcal{S}_0} \frac{\log s_0}{N_j}(1+g)\right).$$

Theoretical Guarantees: Safety and Efficiency

Safe against negative transfer

Under mild conditions, $N_j \asymp \sum_{k=1}^{K} \mathbb{E}(n_{\mathcal{I}_k})$. Since $g \leq \sum_{k=1}^{K} \mathbb{E}(n_{\mathcal{I}_k})\gamma_0^2$, the convergence rate of $\widehat{\boldsymbol{\beta}}_{\text{hiw}}^{(0)}$ is at most of order γ_0^2. Thus, HIW retains the target-only convergence order in the worst case.

Improvement under transferability

If $s_0 = O(1)$, $M_k = h_{\min}$, and $h_{\min} \sum_{k=1}^{K} h_k = O(\log p/n)$ with $n_k \equiv n$, then

$$g = O(\log p), \qquad \|\widehat{\boldsymbol{\beta}}_{\text{hiw}}^{(0)} - \boldsymbol{\beta}^{(0)}\|_2^2 = O_p\left(\sum_{j \in \mathcal{S}_0} \frac{\log p}{N_j}\right) = o_p(\gamma_0^2).$$

HIW can be strictly faster than the target-only estimator.

Data-Fused Projection Matrix

A natural linear imputation matrix would be $(\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\Theta_k})^{-1}\boldsymbol{\Sigma}^{(k)}_{\Theta_k,\bullet}$. In HFS, this cross-covariance block involves unobserved source features, so we use a positive-definite working matrix $\boldsymbol{B}$:

$$\boldsymbol{G}^{(k)} = (\boldsymbol{B}_{\Theta_k,\Theta_k})^{-1}\boldsymbol{B}_{\Theta_k,\bullet}.$$

Data-fused working matrix

$$\boldsymbol{B}^{\mathrm{DF}} = (1-\rho_1-\rho_2)\boldsymbol{I}_p + \rho_1\boldsymbol{\Sigma}^{(0)} + \rho_2\boldsymbol{\Sigma}^{A}, \qquad \rho_1,\rho_2 \geq 0,\ \rho_1+\rho_2 < 1.$$

$$\sigma^{A}_{jl} = \frac{n_0\sigma^{(0)}_{jl} + \sum_{k\in\Xi_{jl}} n_k\sigma^{(k)}_{jl}}{n_0 + \sum_{k\in\Xi_{jl}} n_k}, \qquad \Xi_{jl} = \{k : j,l \in \Theta_k\}.$$

Target-Oriented Projection Refinement

Motivation

The theory shows that $\boldsymbol{G}^{(k)}$ affects the rate through $h_k = \|\boldsymbol{\theta}^{(k)} - \boldsymbol{G}^{(k)}\boldsymbol{\beta}^{(0)}\|_1$.
A good projection should make $\boldsymbol{G}^{(k)}\boldsymbol{\beta}^{(0)}$ close to $\boldsymbol{\theta}^{(k)}$.

Adaptive diagonal scaling

After obtaining $\widehat{\boldsymbol{\beta}}^{(0)}_{\mathrm{hiw}}$, estimate

$$\hat{\boldsymbol{\tau}}^{(k)} = \mathrm{argmin}_{\boldsymbol{\tau}} \left\{ n_k^{-1} \sum_i r_i^2(\boldsymbol{\tau}) + \lambda_\tau \|\boldsymbol{\tau}\|_1 \right\},$$

$$r_i(\boldsymbol{\tau}) = y_i^{(k)} - (\tilde{\boldsymbol{x}}_i^{(k)})^\top \mathrm{diag}(\boldsymbol{\tau})\, \boldsymbol{G}^{(k)} \widehat{\boldsymbol{\beta}}^{(0)}_{\mathrm{hiw}}.$$

The adaptive target-oriented projection is

$$\boldsymbol{G}^{(k)}_{\mathrm{ad}} = \mathrm{diag}(\hat{\boldsymbol{\tau}}^{(k)})\, \boldsymbol{G}^{(k)}.$$

We rerun HIW with $\boldsymbol{G}^{(k)}_{\mathrm{ad}}$ and refer to the resulting estimator as AHIW.

Simulation Study: Settings

Low-dimensional setting

- $(p, n_0, n_k) = (5, 20, 70)$, $K = 10$.
- $\boldsymbol{\beta}^{(0)} = 0.5 \cdot \mathbf{1}_p$.
- K_0 informative sources with $h \in \{0.6, 1\}$.
- X_1 is observed in all sources; X_5 only in one full-feature source.

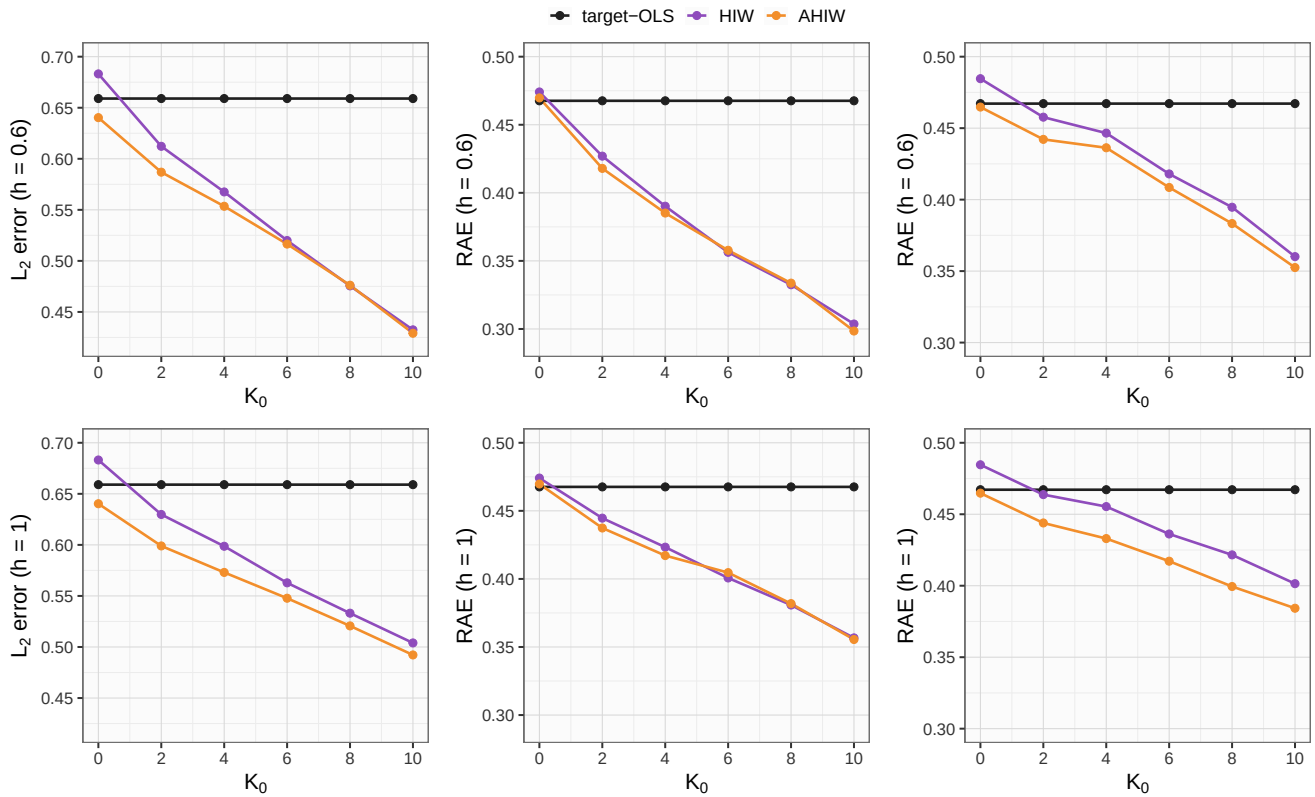
High-dimensional setting

- $(p, n_0, n_k) = (300, 150, 300)$, $s_0 = 10$.
- $\boldsymbol{\beta}^{(0)} = (0.5 \cdot \mathbf{1}_{s_0}^\top, \mathbf{0}^\top)^\top$.
- K_0 informative sources with $h \in \{0.5, 1.5\}$.
- X_1 is observed in all sources; X_{10} only in four sources.

We compare HIW and AHIW with target-only estimators: target-OLS in low dimension and target-LASSO in high dimension.

Simulation Study: Low-Dimensional Results

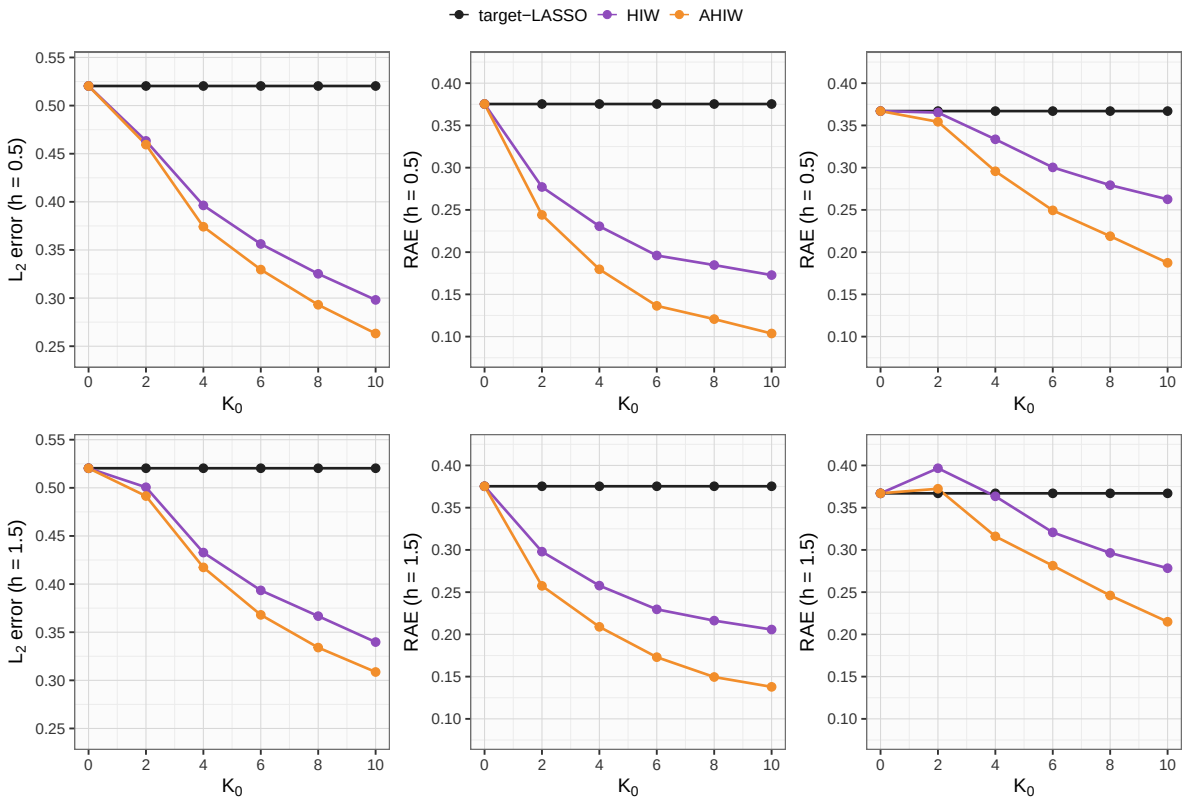
Global and entrywise estimation errors



Lower is better. Columns report global error and two entry-wise RAEs, respectively.

Simulation Study: High-Dimensional Results

Global and entrywise estimation errors



Lower is better. Columns report global error and two entry-wise RAEs, respectively.

Real Data Analysis: GTEx Brain-Tissue Prediction

Dataset Overview

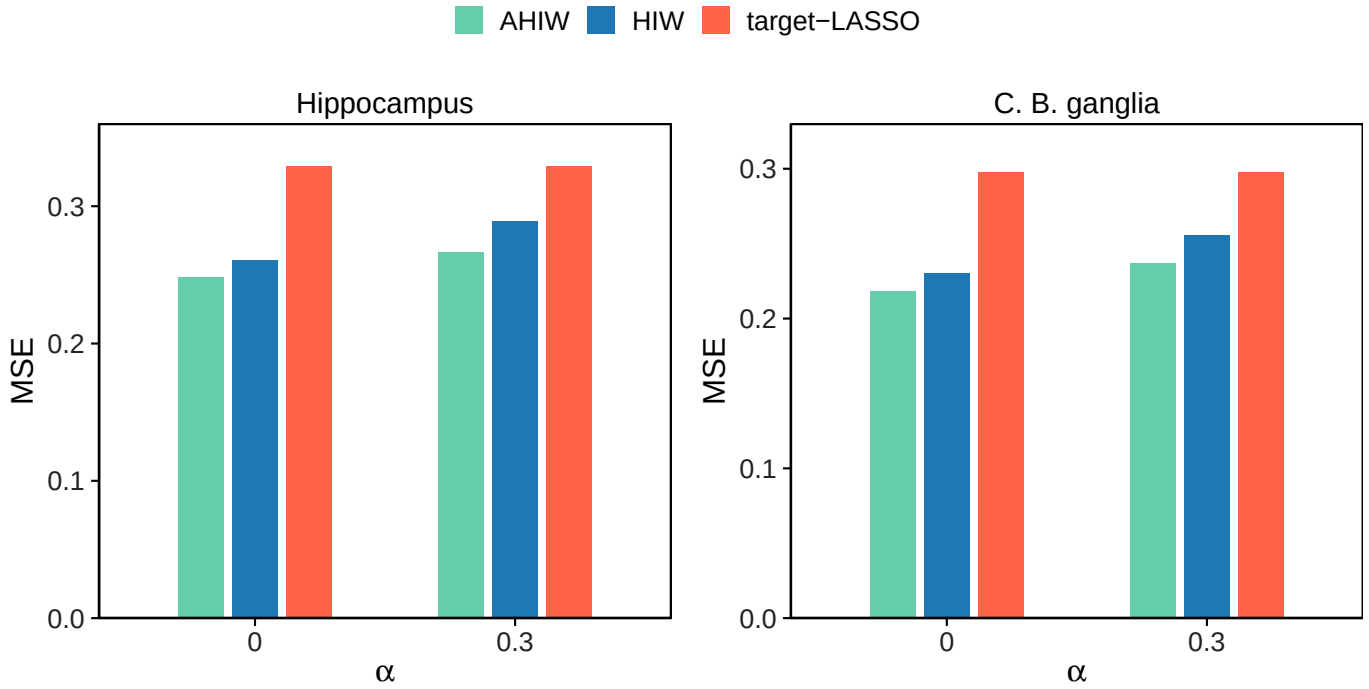
- **Data:** GTEx gene expression data from 49 tissues and 838 donors.
- **Pathway:** GO central nervous system neuron differentiation.
- **Response:** HES5 expression level.
- **Targets:** Hippocampus and Caudate basal ganglia.

Experimental Protocol

- **Sources:** the remaining 11 brain tissues.
- **Covariates:** 153 pathway genes after filtering and excluding HES5.
- **HFS stress test:** additionally remove a proportion $\alpha \in \{0, 0.3\}$ of observed source covariates.
- **Split:** 80% training, 20% testing; 50 replications.
- **Methods:**
  - Baseline: target-LASSO.
  - Ours: HIW and adaptive HIW (AHIW).

Real Data Analysis: Prediction Accuracy

GTEx prediction results (MSE comparison)

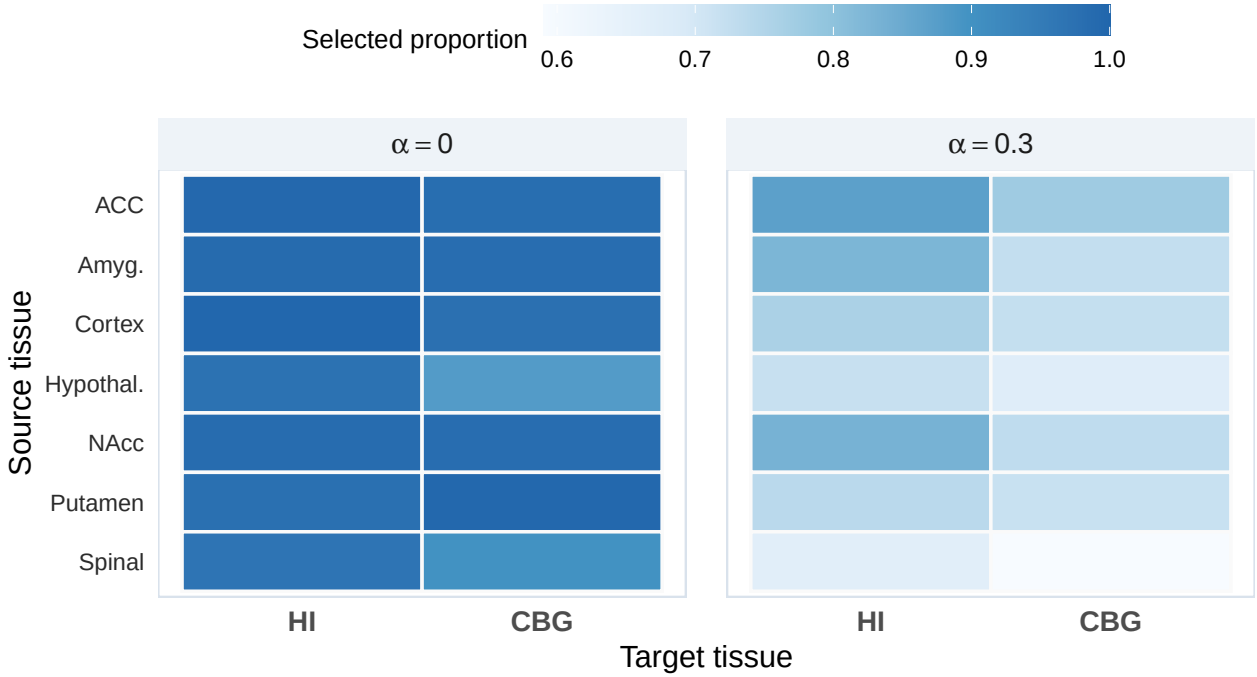


Lower MSE indicates better prediction.

- **Transfer gain:** HIW/AHIW substantially improve over target-LASSO for both target tissues.
- **Robustness:** the gain persists when $\alpha = 0.3$, where source feature mismatch is stronger.

Real Data Analysis: Source Selection

How much source information does AHIW retain?



Darker colors indicate larger retained source-sample proportions.

When feature mismatch is stronger, AHIW can retain fewer source samples.

Thank you!

Q & A